HEALTHCARE DATASET: PROJECT_1

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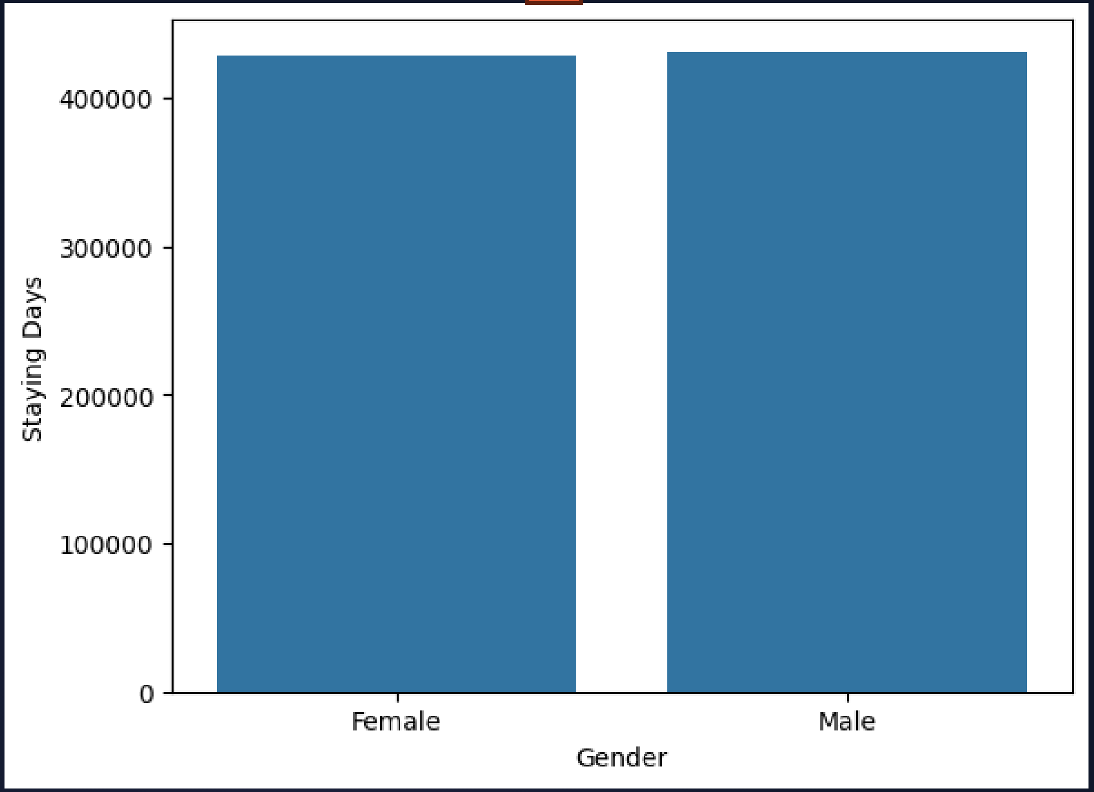
Healthcare Dataset: Exploration & Validation

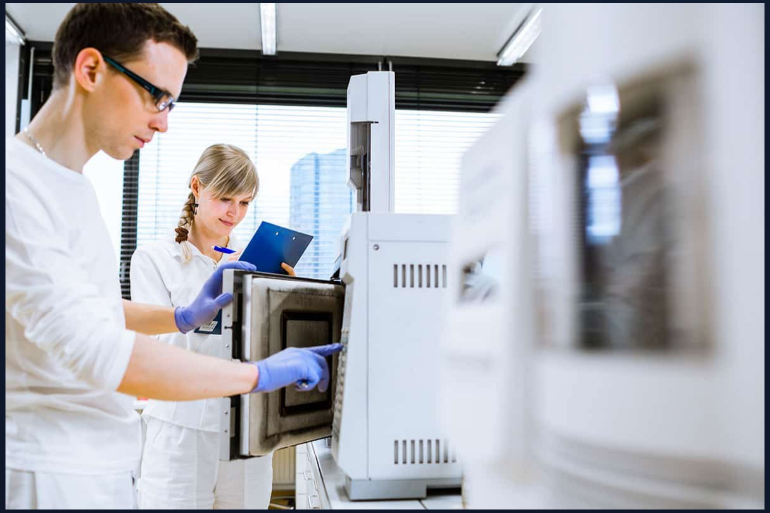
An analysis of a 55,392 patient dataset, examining :

- Gender separation
- Distribution of staying length
- Distribution of billing amount
- Distribution by Age Category
- Billing Amount by Medical Condition
- Medical conditions
- Blood types
- Data integrity

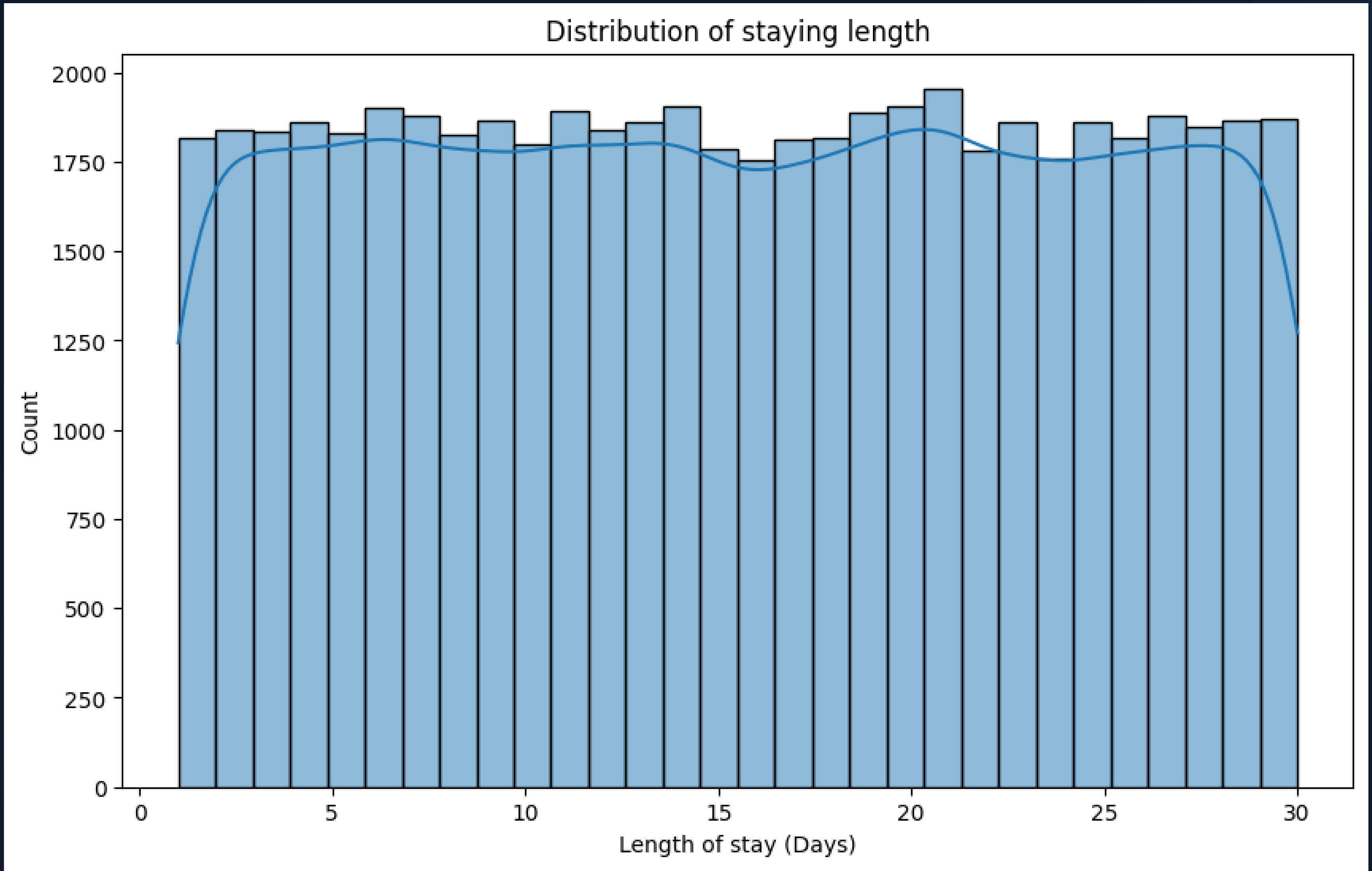
SEPARATION INTO MEN AND WOMEN

	Gender	Staying Days
0	Male	430862
1	Female	428200



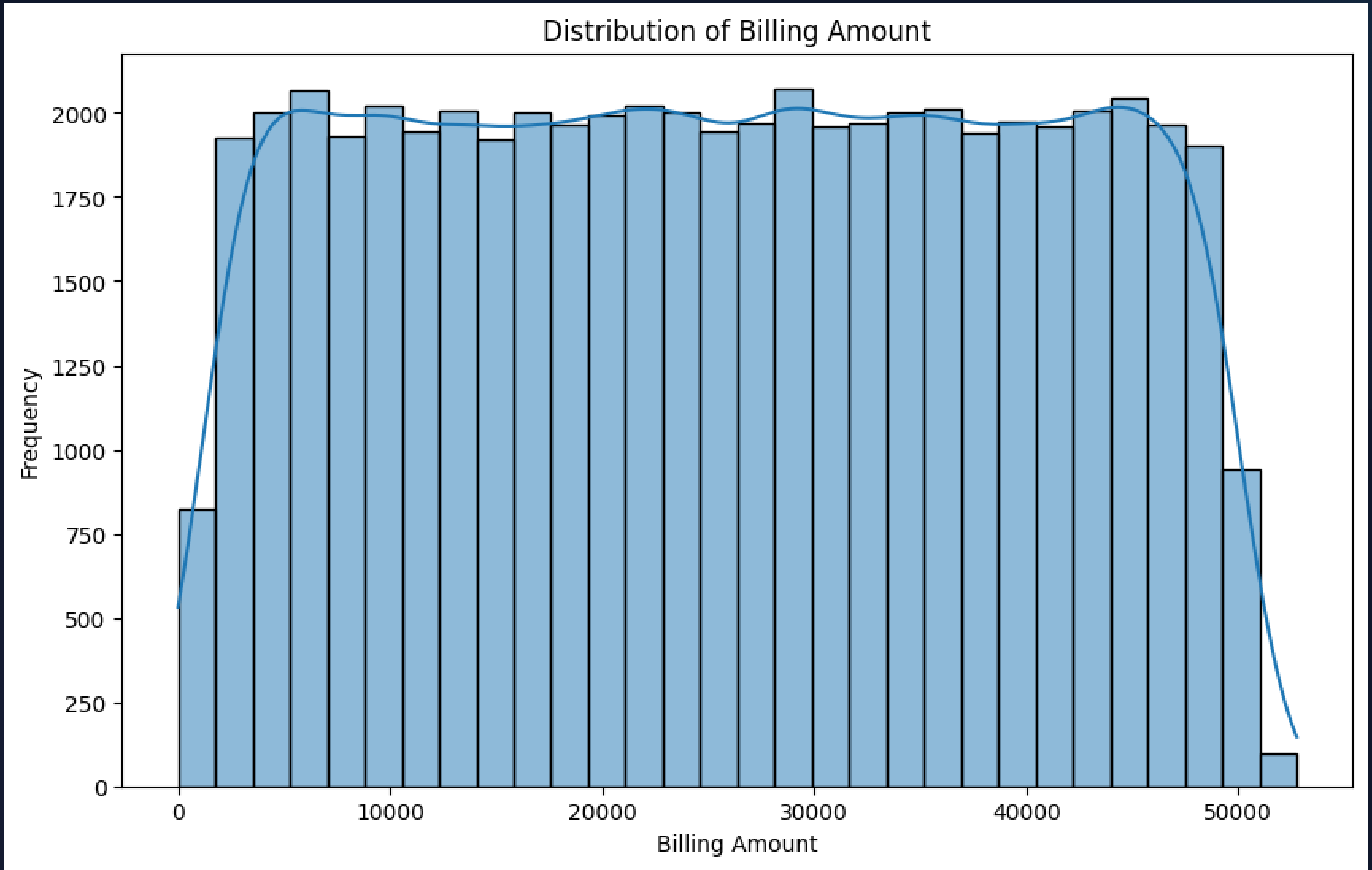


DISTRIBUTION OF STAYING LENGTH



- According to the graph the data are distributed uniformly.
- Maximum length of stay is 30 days.
- Minimum length of stay is 1 day (the patient had a simple service).

DISTRIBUTION OF BILLING AMOUNT

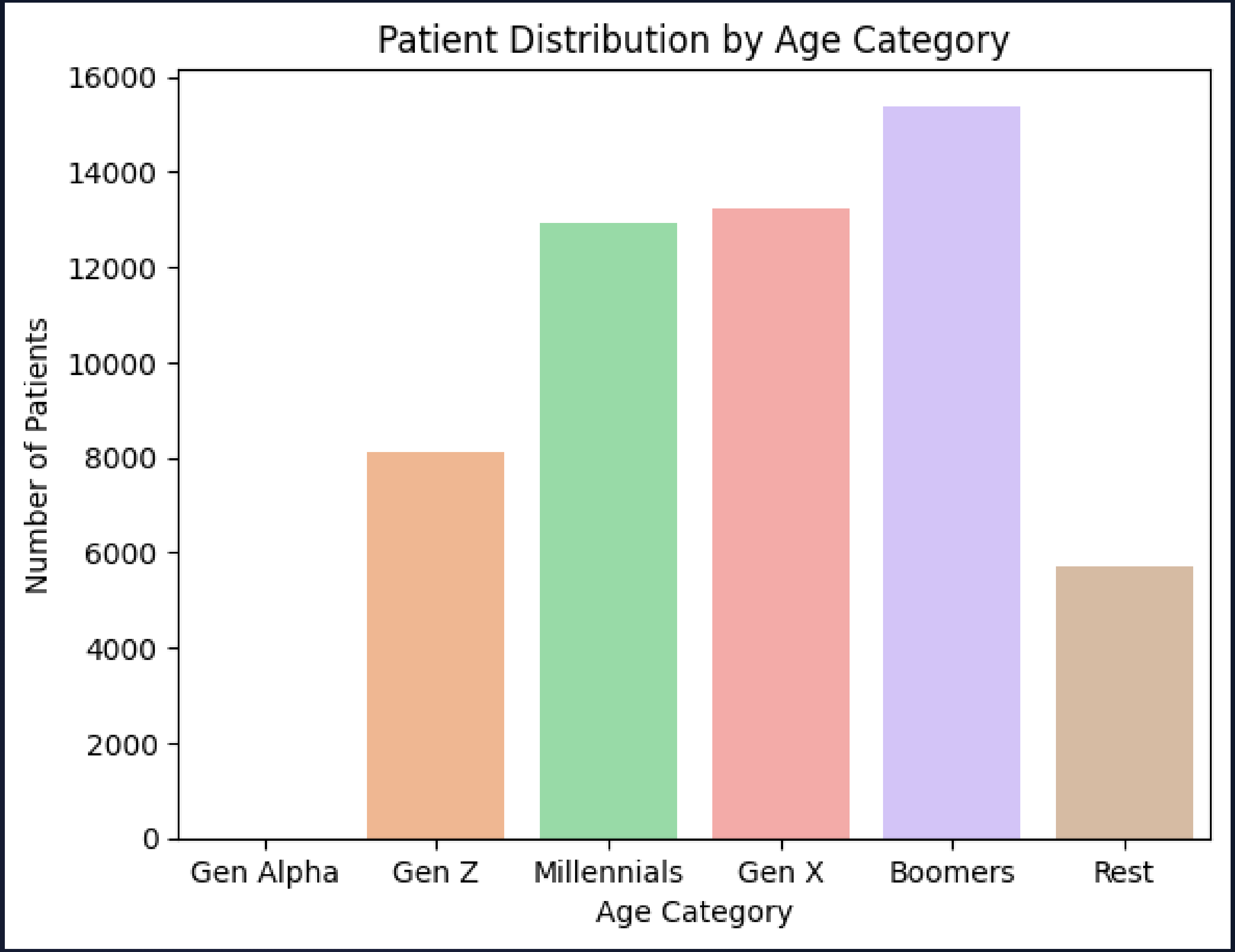


- Max billing amount 52.765 dollar.
- Minimum billing amount 9.20 dollars (< 1 days staying).

TOP 20 CUSTOMERS BY LENGTH OF STAY

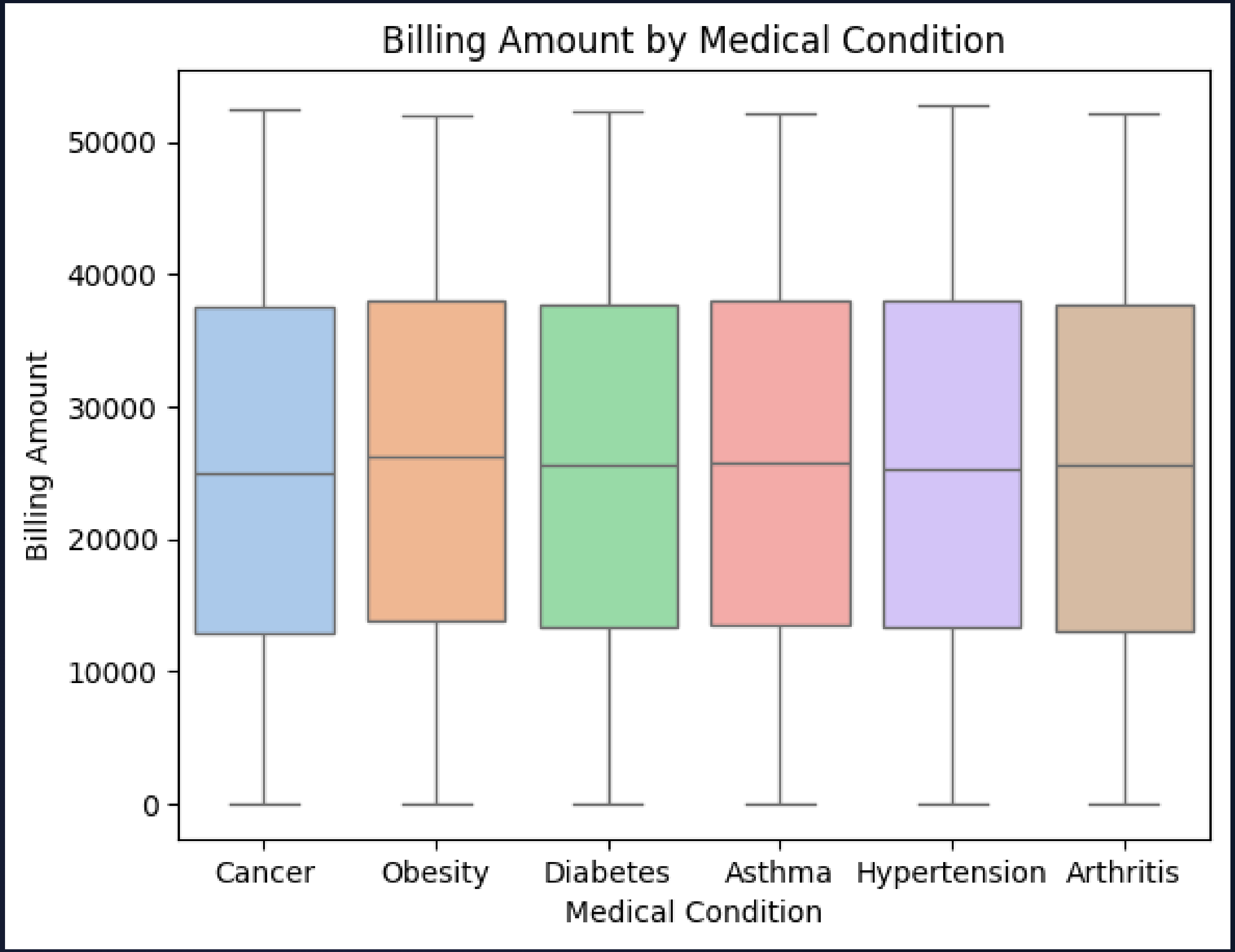
	Name	Staying Days	Billing Amount
0	MICHAEL SMITH	376 days	660173.089022
1	MICHAEL WILLIAMS	318 days	555240.698319
2	JAMES GARCIA	307 days	346601.993435
3	JAMES WILLIAMS	298 days	471240.445227
4	JAMES SMITH	296 days	468220.237192
5	JAMES BROWN	289 days	452411.246237
6	JOHN SMITH	266 days	373678.436266
7	KIMBERLY SMITH	265 days	308210.814286
8	MATTHEW JONES	247 days	327497.857234
9	ROBERT SMITH	245 days	573407.937591
10	DAVID JOHNSON	239 days	457290.174635
11	THOMAS SMITH	237 days	376672.146342
12	CHRISTOPHER SMITH	234 days	400333.251755
13	MATTHEW SMITH	228 days	268883.354186
14	DAVID SMITH	228 days	308166.113499
15	BRIAN SMITH	220 days	311323.295963
16	DANIEL SMITH	218 days	243125.307905
17	JENNIFER JONES	216 days	384280.262095
18	ROBERT WILLIAMS	215 days	318611.531787
19	JAMES DAVIS	215 days	319207.624626

PATIENT DISTRIBUTION BY AGE CATEGORY



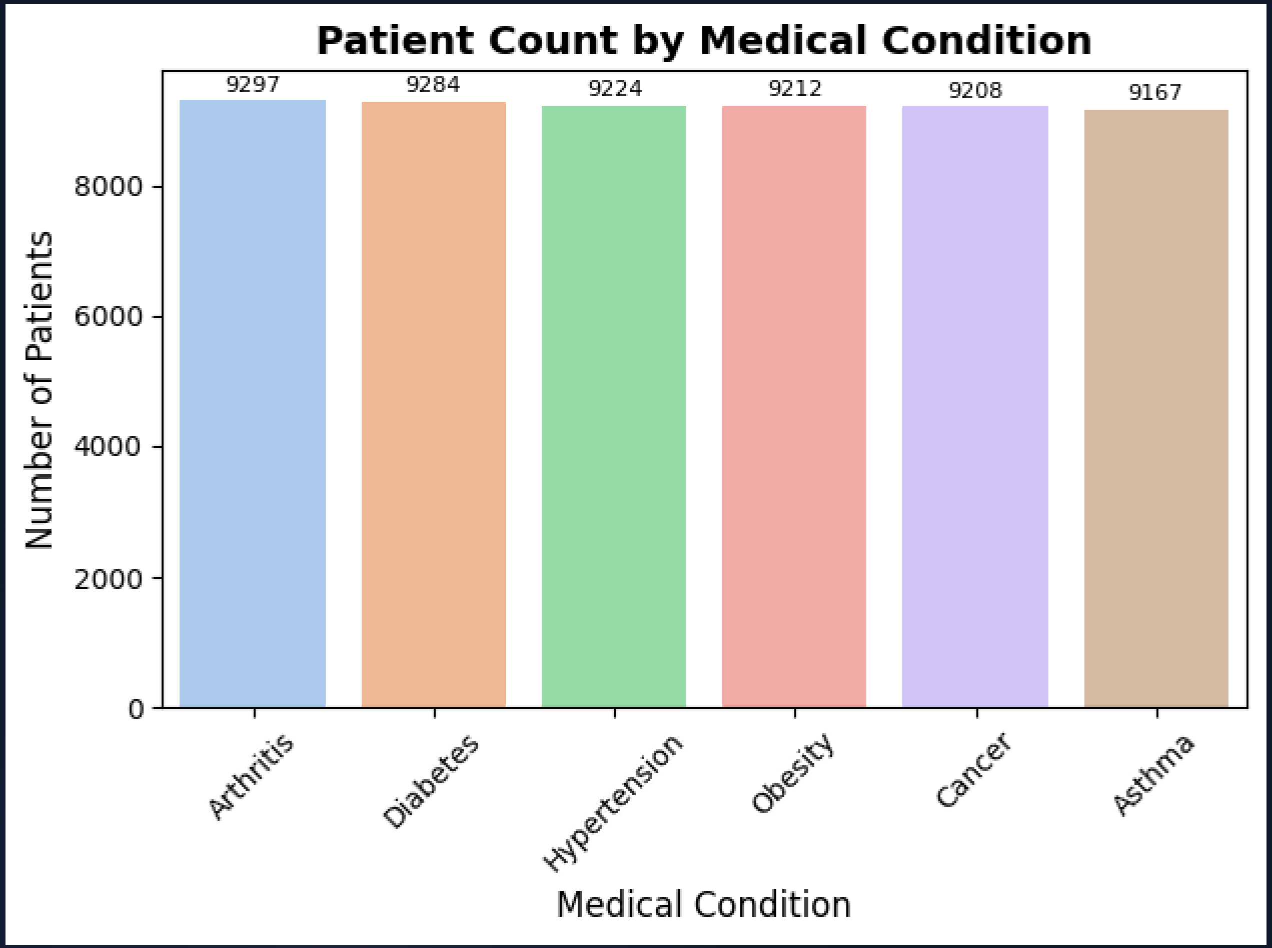
- Boomers and Gen X have the first class of patients
- Gen Alpha no contact with hospitals

PATIENT DISTRIBUTION BY AGE CATEGORY



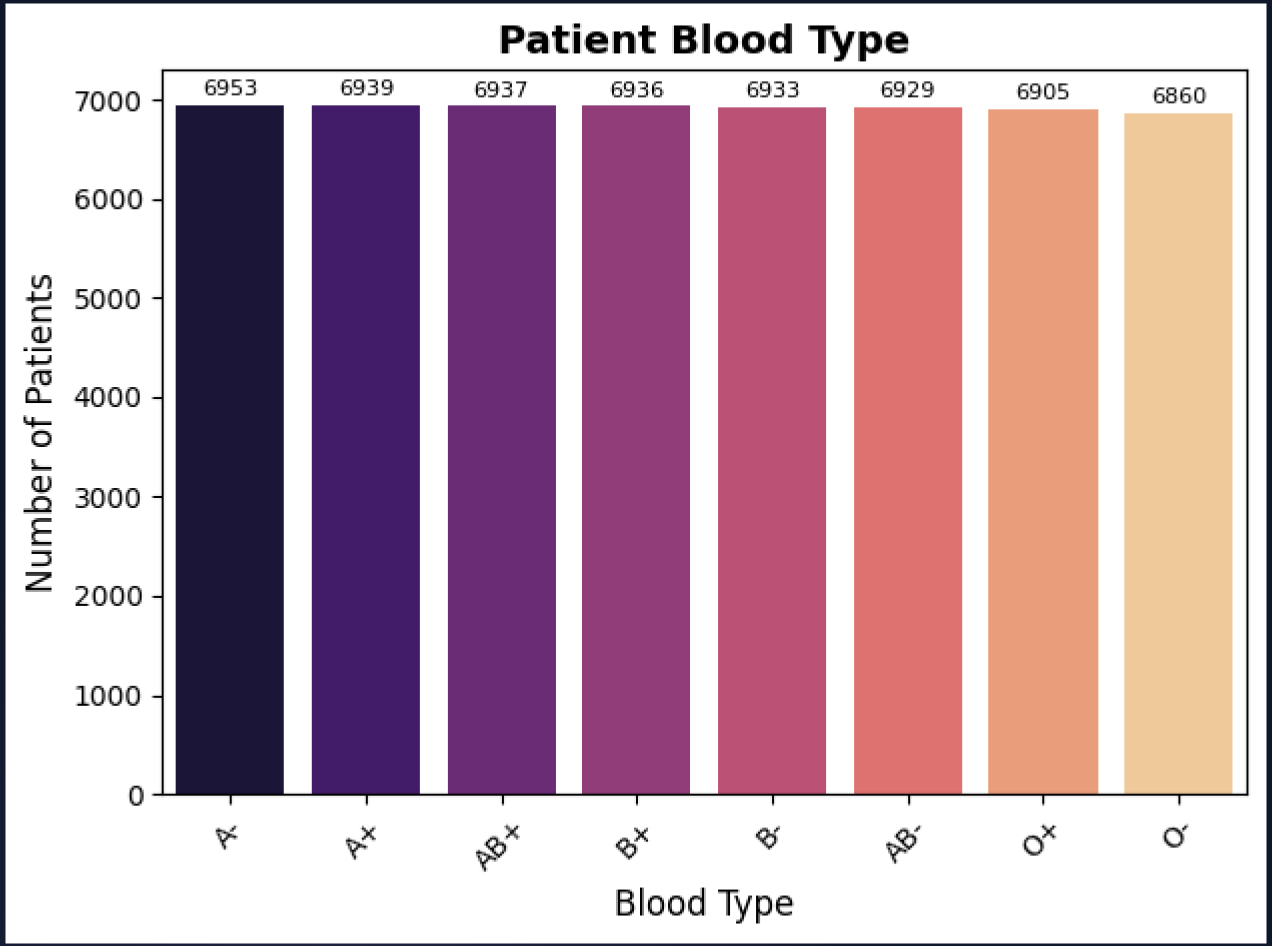
- All medical conditions have slight differences in payments.

MOST COMMON MEDICAL CONDITIONS



- Arthritis is technically the most common (9.297 cases).
- However, the distribution across all six conditions is remarkably even, with each accounting for roughly 16% of the dataset.

MOST COMMON BLOOD TYPES



- Blood type A- is the most frequent.
- Like medical conditions, every blood type (including rare ones like AB-) constitutes approximately 12.5% of the patient population.

DATA INTEGRITY-CONCLUSION

This is a synthetic dataset: Generated using uniform distributions (likely via Python's Faker library), this data was created for practice, not real-world insights.

Excellent for coding practice: It remains highly valuable for practicing data manipulation, cleaning, EDA, and visualization techniques using Pandas and Matplotlib.